Evaluating Two-Input Transfer Function Analysis of Cerebral Hemodynamics

Known-truth simulation framework and a 250-family proof of concept

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How this presentation is being used

PURPOSE

This presentation is organized like a developing paper rather than a short conference talk. The early sections establish the signal-processing and physiological reasoning step by step. The later sections define the simulation, show how estimates are compared with known truth, and then introduce the first proof-of-concept results.

The goal is not to make the two-input model look better by default. The goal is to identify the conditions in which jointly estimating MAP and $P_{\mathrm{ET}}CO_2$ recovers their separate contributions to CBFV more accurately than treating each input alone.

SECTION 1

Introduction and Motivation

CBFV is influenced by more than arterial pressure. That creates a reason to consider two inputs, but it does not guarantee that a larger model will be more accurate.

Cerebral blood flow is regulated despite changing pressure

INTRODUCTION AND MOTIVATION

The brain requires continuous blood flow even though systemic arterial pressure changes from moment to moment. Cerebral autoregulation describes part of the vascular response that limits how strongly these pressure fluctuations alter cerebral perfusion. Dynamic cerebral autoregulation focuses on changes occurring over seconds to minutes and is commonly studied by treating MAP as an input and CBFV as an output.

Transfer function analysis is useful because it describes this relationship at each frequency. Gain represents the size of the response and phase represents its timing. These measures describe a dynamic relationship; they do not prove that one signal is the only physiological cause of the other.

Methodological basis: Zhang et al. (1998), doi:10.1152/ajpheart.1998.274.1.H233; Claassen et al. (2016), doi:10.1177/0271678X15626425; Panerai et al. (2023), doi:10.1177/0271678X221119760.

$P_{\mathrm{ET}}CO_2$ creates a second measured influence on CBFV

INTRODUCTION AND MOTIVATION

Carbon dioxide changes cerebrovascular tone and can influence CBFV in frequency ranges that overlap with the ranges used for autoregulation analysis. End-tidal carbon dioxide is not identical to arterial carbon dioxide, but it provides a continuous practical measurement during resting recordings.

If MAP and $P_{\mathrm{ET}}CO_2$ share fluctuations, a MAP-only transfer function can contain variation associated with both pathways. The same issue can occur in the reverse direction for a $P_{\mathrm{ET}}CO_2$-only estimate. Separate SISO estimates remain valid pairwise descriptions, but they should not automatically be interpreted as fully separated physiological pathways.

CO_2 influence and multivariable motivation: Panerai et al. (1999), doi:10.1088/0967-3334/20/3/304; Panerai et al. (2000), doi:10.1109/10.827312; Peng et al. (2008), doi:10.1007/s10439-007-9412-9.

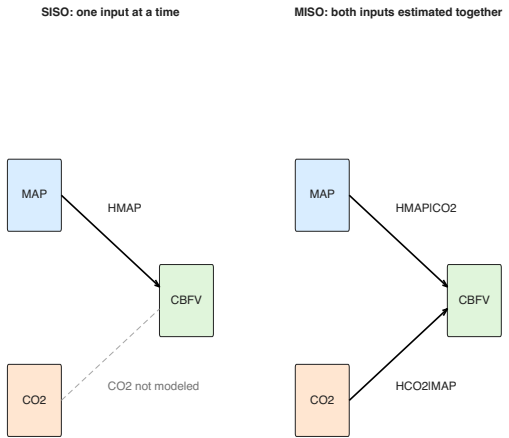
The study asks a conditional question

INTRODUCTION AND MOTIVATION

A two-input model provides a mathematical way to separate shared information, but it must estimate two pathway coefficients from the same finite recording. When the inputs are difficult to distinguish, the joint spectral system can become sensitive to small estimation errors.

The useful question is therefore: *under which combinations of input relationship, pathway strength, recording quality, and estimator settings does MISO recover the known pathways more accurately than separate SISO models?* Human recordings cannot answer this accuracy question by themselves because the true physiological transfer functions are not directly observable.

SISO and MISO answer related but different questions



Conceptual method diagram generated by the MATLAB project; it does not present numerical evidence.

SECTION 2

From Physiological Signals to Spectral Estimates

Transfer functions are estimated from auto-spectra and cross-spectra. The foundation begins with how a finite time signal becomes a stable spectral estimate.

A transfer function maps an input to an output

SIGNAL-PROCESSING BACKGROUND

If $x(t)$ is an input and $y(t)$ is an output, a simple static analogy is $y = mx$: the slope m describes how strongly the input is represented in the output. A transfer function plays a similar role but can have a different value at every frequency and can also describe timing.

$$Y(f) = H(f)X(f)$$

The magnitude $|H(f)|$ is gain, while the angle $\angle H(f)$ is phase. Together they describe the size and timing of the output response at frequency f.

The frequency domain reorganizes the same signal

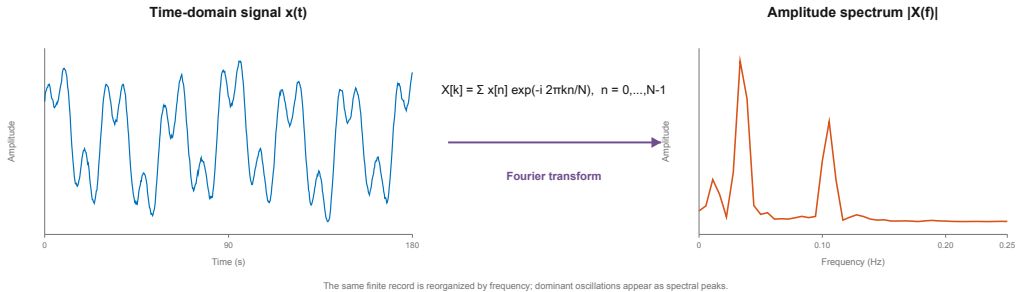
SIGNAL-PROCESSING BACKGROUND

MAP, $P_{\mathrm{ET}}CO_2$, and CBFV are first recorded in the time domain. Each sample describes the signal at one time, which is useful for seeing trends, artifacts, noise, and timing relationships. Several oscillations can overlap in a single time trace, however, making their individual contributions hard to see.

The Fourier transform reorganizes that finite record into discrete frequency values. Slow oscillations appear at low frequencies and faster oscillations at higher frequencies. Each Fourier value is generally complex: its magnitude describes the strength of that component and its angle describes phase relative to a reference sinusoid.

The Fourier transform exposes the frequencies inside the signal

SIGNAL-PROCESSING BACKGROUND



Generated from a sampled signal by `figure_generators/generate_method_figures.py`; the peaks correspond to oscillations present in the time trace.

A complex spectrum stores both amplitude and phase

SIGNAL-PROCESSING BACKGROUND

Every Fourier value can be written in polar form. Its magnitude gives the strength of that frequency component, while its angle gives the component's phase relative to the time origin.

$$X(f) = |X(f)|e^{i\angle X(f)}.$$

A power spectrum is based on $|X(f)|^2$ and is therefore real and nonnegative. It describes how the signal's power is distributed across frequency, but it no longer retains the phase of the original Fourier value. A cross-spectrum uses two signals and remains complex, so it can retain their relative phase.

Welch's method creates repeated estimates inside one record

SIGNAL-PROCESSING BACKGROUND

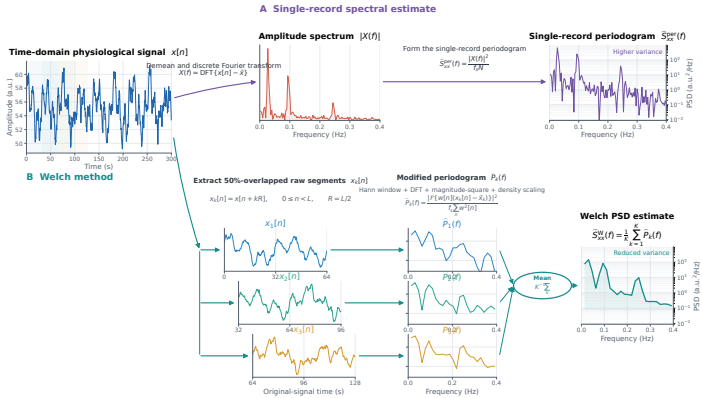
Welch's method divides the original record into shorter, often overlapping segments. Every segment contains the actual samples from its corresponding time interval; overlapping segments share some of those samples rather than creating different versions of the signal.

Each segment is multiplied by a window such as a Hann window, transformed, and converted into a segment-level spectral estimate. Because all segments use the same frequency grid, their periodograms or cross-periodograms can be averaged frequency by frequency. This reduces variance but creates a tradeoff: shorter windows provide more averages, whereas longer windows provide finer frequency resolution.

Spectral-estimation basis: Welch (1967), doi:10.1109/TAU.1967.1161901; current cerebral-TFA recommendations: Panerai et al. (2023), doi:10.1177/0271678X221119760.

Two routes from one time signal to a spectral estimate

SIGNAL-PROCESSING BACKGROUND



The lower route shows overlapping samples from one signal, segment-level modified periodograms, and their frequency-by-frequency average.

Auto-spectra describe power within one signal

SIGNAL-PROCESSING BACKGROUND

For segment m, let $X_m(f)$ be the Fourier transform of the windowed signal. The segment power estimate contains the product of the complex value and its own complex conjugate. Welch's estimate averages these products across the M available segments.

$$\widehat{S}_{xx}(f) = \frac{1}{M} \sum_{m=1}^{M} X_m^*(f) X_m(f) = \frac{1}{M} \sum_{m=1}^{M} |X_m(f)|^2$$

The implementation also includes the sampling-frequency and window-energy normalization needed for a correctly scaled power spectral density.

Cross-spectra retain the relationship between two signals

SIGNAL-PROCESSING BACKGROUND

For an input X and output V, the cross-spectrum uses the conjugate of the first signal and the Fourier value of the second signal. Its magnitude describes shared frequency content, while its angle contains the relative phase.

$$\widehat{S}_{xV}(f) = \frac{1}{M} \sum_{m=1}^{M} X_m^*(f) V_m(f)$$

The ordering must be stated because reversing the two signals conjugates the cross-spectrum and changes the sign of phase.

The code implementation follows the spectral definitions

CODE IMPLEMENTATION

```
Sxx = 0;
SxV = 0;
for m = 1:M
    X = fft(window .* xSegment(:,m));
    V = fft(window .* vSegment(:,m));
    Sxx = Sxx + conj(X).*X;
    SxV = SxV + conj(X).*V;
end
Sxx = Sxx/M;
SxV = SxV/M;
```

Conceptual implementation. Production functions additionally apply PSD normalization, smoothing, and input checks.

SECTION 3

Conventional SISO Transfer-Function Analysis

The SISO estimator describes one selected input-output relationship at a time.

A dynamic time-domain response becomes multiplication by frequency

SISO TRANSFER-FUNCTION ANALYSIS

Let $x(t)$ be one selected input and let $v(t)$ be CBFV. The output at one time can depend on the current and previous values of the input. The impulse response $h(t)$ describes those delayed and distributed effects, while $e(t)$ contains output variation not explained by this SISO model.

$$v(t) = (h * x)(t) + e(t),$$

$$V(f) = H(f)X(f) + E(f).$$

The Fourier transform changes convolution into a frequency-by-frequency multiplication. This is why transfer function analysis can estimate the dynamic relationship separately at each frequency.

Direct FFT division makes the unexplained term visible

SISO TRANSFER-FUNCTION ANALYSIS

Dividing the frequency-domain equation by $X(f)$ gives useful intuition, but it does not isolate the true pathway when the finite record contains unexplained output variation.

$$\frac{V(f)}{X(f)} = H(f) + \frac{E(f)}{X(f)}.$$

If $X(f)$ is small, even modest unexplained variation can dominate the ratio. A single FFT ratio also depends heavily on the particular finite realization. The estimator therefore creates products that can be averaged across Welch segments rather than dividing two individual Fourier values.

Multiplying by the input conjugate creates spectral products

SISO TRANSFER-FUNCTION ANALYSIS

Multiply both sides of $V(f) = H(f)X(f) + E(f)$ by the complex conjugate of the input. The pathway is now multiplied by input power rather than appearing below an individual Fourier value.

$$X^*(f)V(f) = H(f)X^*(f)X(f) + X^*(f)E(f).$$

The first product becomes the input-output cross-spectrum after averaging. The second product becomes the input auto-spectrum. The remaining product describes any systematic relationship between the input and the unexplained output term.

Averaging isolates the modeled relationship in expectation

SISO TRANSFER-FUNCTION ANALYSIS

Taking the expectation, which Welch averaging approximates using finite segments, gives

$$\mathbb{E}[X^* V] = H(f)\mathbb{E}[X^* X] + \mathbb{E}[X^* E].$$

If the unexplained output term is not systematically related to the input, then $\mathbb{E}[X^* E] \approx 0$. The remaining terms are the cross-spectrum and auto-spectrum:

$$S_{xV}(f) = H(f)S_{xx}(f).$$

The SISO estimator follows by solving the spectral equation

SISO TRANSFER-FUNCTION ANALYSIS

Solving $S_{xV} = HS_{xx}$ for the pathway gives the familiar estimator. Hats indicate spectra and transfer functions estimated from a finite observation.

$$\widehat{H}_x^{\mathrm{SISO}}(f) = \frac{\widehat{S}_{xV}(f)}{\widehat{S}_{xx}(f)}, \qquad \widehat{G}(f) = |\widehat{H}(f)|, \qquad \widehat{\phi}(f) = \angle \widehat{H}(f).$$

The numerator describes how the input and CBFV vary together. The denominator scales that relationship by the input power available at the same frequency. Gain and phase are calculated only after the complex transfer function has been estimated.

The SISO code is the mathematical estimator written directly

CODE IMPLEMENTATION

```
% Sxx and SxV were averaged across Welch segments
H = SxV ./ Sxx;

gain = abs(H);
phase = angle(H);
```

The element-wise division is performed frequency by frequency. The production implementation also checks finite values, preserves the selected frequency range, and applies the chosen phase-unwrapping method, but those steps do not change the estimator shown above.

Coherence describes consistency, not physiological truth

SISO TRANSFER-FUNCTION ANALYSIS

Magnitude-squared coherence compares the input-output cross-spectrum with the power present in both signals.

$$\gamma_{xV}^2(f) = \frac{|S_{xV}(f)|^2}{S_{xx}(f)S_{VV}(f)}, \qquad 0 \leq \gamma_{xV}^2(f) \leq 1$$

High coherence means that a consistent linear pairwise relationship is present at that frequency. It does not prove causality, and it does not show that a pairwise SISO estimate equals the underlying MAP or $P_{\mathrm{ET}}CO_2$ pathway after the other measured input is considered.

A SISO estimate is a pairwise description

SISO TRANSFER-FUNCTION ANALYSIS

The MAP SISO transfer function answers how MAP and CBFV are related when MAP is the only modeled input. The $P_{\mathrm{ET}}CO_2$ SISO transfer function answers the corresponding pairwise question for $P_{\mathrm{ET}}CO_2$ and CBFV.

Neither calculation is mathematically wrong. The limitation appears when the result is interpreted as a uniquely separated pathway even though the other measured input is related to both the selected input and the output. This distinction motivates the MISO derivation rather than an assumption that a larger model must be better.

SECTION 4

Why a SISO Estimate Can Contain the Second Pathway

The shared-input term can be derived directly from the true two-input system.

Begin with a true two-input linear system

WHY A SECOND INPUT MATTERS

Let M represent MAP, C represent $P_{\mathrm{ET}}CO_2$, and V represent CBFV. Each input has its own true transfer function, while E contains output variation not explained by the two measured inputs.

$$V(f) = H_M(f)M(f) + H_C(f)C(f) + E(f)$$

For the spectral derivation, the unexplained term is assumed to be uncorrelated with both measured inputs. The known-truth simulation later adds finite-record variation and measurement noise while retaining known H_M and H_C.

Two-input cerebral-hemodynamic models: Panerai et al. (2000), doi:10.1109/10.827312; Peng et al. (2008), doi:10.1007/s10439-007-9412-9; Henley et al. (2015), PMCID: PMC4756910.

Relating MAP to the output keeps both pathway terms

WHY A SECOND INPUT MATTERS

Multiply the two-input system by $M^*(f)$ and take the expectation. The MAP-to-output cross-spectrum contains the MAP pathway multiplied by MAP power, but it also contains a second term whenever MAP and $P_{\mathrm{ET}}CO_2$ have a nonzero cross-spectrum.

$$S_{MV}(f) = H_M(f)S_{MM}(f) + H_C(f)S_{MC}(f)$$

The second term is not noise. It represents the true $P_{\mathrm{ET}}CO_2$ pathway carried into a pairwise MAP-output relationship through the relationship between the two inputs.

The MAP SISO estimate can include a $P_{\mathrm{ET}}CO_2$ term

WHY A SECOND INPUT MATTERS

Substituting the two-input cross-spectrum into the SISO estimator gives a direct expression for what MAP-only TFA estimates.

$$H_M^{\mathrm{SISO}}(f) = H_M(f) + H_C(f)\frac{S_{MC}(f)}{S_{MM}(f)}$$

The first term is the MAP pathway. The second depends on the $P_{\mathrm{ET}}CO_2$ pathway and on the complex relationship between the two inputs. Its magnitude and phase can change across frequency, so it cannot generally be removed with one scalar correction.

The omitted-input term follows directly from the displayed linear system; its physiological motivation is supported by Panerai et al. (2000) and Peng et al. (2008).

The reverse occurs for the $P_{\mathrm{ET}}CO_2$ SISO estimate

WHY A SECOND INPUT MATTERS

Repeating the same derivation with $P_{\mathrm{ET}}CO_2$ as the selected input produces the corresponding shared-input term.

$$H_C^{\mathrm{SISO}}(f) = H_C(f) + H_M(f)\frac{S_{CM}(f)}{S_{CC}(f)}$$

If $S_{MC}(f) = 0$, this specific mixing term disappears at that frequency. This special case explains why MISO should not be expected to improve every record equally: when the inputs are nearly independent, there may be little SISO mixing to remove.

SECTION 5

The Joint MISO Model

MISO solves both pathway coefficients together using the complete two-input spectral matrix.

The two pathway estimates are solved together

THE MISO MODEL

At each frequency, the diagonal terms contain MAP and $P_{\mathrm{ET}}CO_2$ power, the off-diagonal terms contain their cross-spectrum, and the right-side vector connects each input to CBFV.

$$\underbrace{\begin{bmatrix} S_{MM} & S_{MC} \\ S_{CM} & S_{CC} \end{bmatrix}}_{\mathsf{S}_{xx}(f)} \underbrace{\begin{bmatrix} H_M \\ H_C \end{bmatrix}}_{\mathsf{H}(f)} = \underbrace{\begin{bmatrix} S_{MV} \\ S_{CV} \end{bmatrix}}_{\mathsf{s}_{xV}(f)}$$

The system uses both unique and shared input information. The resulting pathway coefficients are conditional estimates: each describes one input-output relationship while accounting for the other measured input.

Multivariable precedent: Panerai et al. (2000), doi:10.1109/10.827312; Peng et al. (2008), doi:10.1007/s10439-007-9412-9; Kostoglou et al. (2024), doi:10.1177/0271678X241249276.

The implementation uses a linear-system solve

THE MISO MODEL

The compact estimator is written with the input spectral matrix $\widehat{\mathrm{S}}_{xx}$ and input-output cross-spectrum vector $\widehat{\mathrm{s}}_{xV}$. The lowercase bold s indicates a vector, whereas S_{xx} is a two-by-two matrix.

$$\widehat{\mathrm{H}}(f) = \widehat{\mathrm{S}}_{xx}(f) \backslash \widehat{\mathrm{s}}_{xV}(f)$$

The backslash denotes solving the two-by-two linear system. It has the same mathematical purpose as multiplying by an inverse, but numerical software solves the equations directly because that is more stable and avoids forming an unnecessary explicit inverse.

The MISO code implements the displayed matrix equation

CODE IMPLEMENTATION

```
Sxx = [SMM, SMC;
       SCM, SCC];

sxV = [SMV;
       SCV];

H = Sxx \ sxV;
HM = H(1);
HC = H(2);
```

Conceptual implementation of $S_{xx}H = s_{xV}$. Production code additionally checks Hermitian symmetry, positive input power, conditioning, and optional ridge regularization.

The two-by-two solution reveals a shared denominator

THE MISO MODEL

Expanding the linear-system solution shows where the determinant enters. Both pathway coefficients contain the same denominator:

$$H_M(f) = \frac{S_{CC}S_{MV} - S_{MC}S_{CV}}{D(f)},$$

$$H_C(f) = \frac{S_{MM}S_{CV} - S_{CM}S_{MV}}{D(f)},$$

$$D(f) = S_{MM}S_{CC} - S_{MC}S_{CM}.$$

The numerators combine the two input-output relationships. The denominator determines whether those relationships contain enough distinct input information to separate the two pathways.

Identifiability means that the two pathways can be distinguished

THE MISO MODEL

The joint model requires some MAP variation that cannot be predicted from $P_{\mathrm{ET}}CO_2$ and some $P_{\mathrm{ET}}CO_2$ variation that cannot be predicted from MAP. Because a spectral matrix is Hermitian, $S_{CM} = S_{MC}^{*}$, so the denominator from the previous slide becomes

$$D(f) = S_{MM}(f)S_{CC}(f) - |S_{MC}(f)|^2$$

When $D(f)$ approaches zero, the two equations contain nearly redundant information. Dividing by this small denominator can amplify finite-record spectral errors and produce large changes in the separated pathway coefficients.

Conditioning context: Panerai et al. (2000); Peng et al. (2008); Kostoglou et al. (2024).

Input coherence connects the physiology to matrix conditioning

THE MISO MODEL

Magnitude-squared input coherence can be substituted into the determinant:

$$D(f) = S_{MM}(f)S_{CC}(f)\left[1 - \gamma^2_{MC}(f)\right].$$

As input coherence approaches one, the normalized determinant approaches zero. This does not mean that any shared input variation invalidates MISO. It means that the benefit of removing SISO mixing must be weighed against the declining amount of independent information available to separate the two coefficients.

Mean removal and spectral scaling solve different problems

THE MISO MODEL

Empirical preprocessing removes each time-domain mean; the primary analysis does not linearly detrend. Mean removal eliminates only the constant, zero-frequency baseline. It does not equalize MAP and $P_{\mathrm{ET}}CO_2$ units, variance, or frequency-specific spectral power.

$$\mathsf{D}(f) = \operatorname{diag}\left(\sqrt{S_{MM}(f)}, \sqrt{S_{CC}(f)}\right).$$

$$\widetilde{\mathsf{S}}_{xx}(f) = \mathsf{D}^{-1}\mathsf{S}_{xx}\mathsf{D}^{-1} = \begin{bmatrix} 1 & \rho(f) \\ \rho^*(f) & 1 \end{bmatrix}, \qquad |\rho(f)|^2 = \gamma^2_{MC}(f).$$

This per-frequency scaling is used only for conditioning diagnostics and ridge sensitivity, preventing input units from dominating those comparisons. The ordinary unregularized coefficients remain in their physical units.

SECTION 6

Input Coherence, PSD Overlap, and Excitation

These ideas are related, but each describes a different part of the information available to a two-input model.

Input coherence describes shared fluctuations

INPUT RELATIONSHIPS AND EXCITATION

MAP-$P_{\mathrm{ET}}CO_2$ coherence measures how consistently the two inputs are linearly related at each frequency. High coherence means that their fluctuations are strongly connected there; low coherence means that more of the observed variation is independent.

Coherence concerns the relationship between the two inputs. It does not say whether either input contains enough power to estimate its pathway precisely, and it is not interchangeable with the amount of overlap between their power spectral density shapes.

PSD overlap describes shared frequency support

INPUT RELATIONSHIPS AND EXCITATION

PSD-shape overlap asks whether MAP and $P_{\mathrm{ET}}CO_2$ contain power in similar frequency ranges. Two inputs can occupy the same broad frequencies without having strongly coherent sample-by-sample fluctuations. Conversely, a pathway cannot be estimated where its input has almost no measurable power, even if a nominal relationship exists elsewhere.

The simulation therefore varies a spectral-similarity control separately from target coherence. Its realized PSD-shape overlap is measured from a long clean record and later grouped into prespecified fixed intervals for descriptive operating maps.

Excitation separates input strength from pathway strength

INPUT RELATIONSHIPS AND EXCITATION

Input excitation describes how strongly an input fluctuates in the analyzed frequency range. Pathway gain describes how strongly CBFV responds to those fluctuations. A weak $P_{\mathrm{ET}}CO_2$ input and a weak $P_{\mathrm{ET}}CO_2$-to-CBFV pathway can both produce a small output contribution, but they are different physiological and estimation conditions.

The simulation varies the $P_{\mathrm{ET}}CO_2$-to-MAP fluctuation standard-deviation ratio and the $P_{\mathrm{ET}}CO_2$-to-MAP pathway band-gain ratio separately so that their joint effect can be examined rather than compressed into one control.

The expected MISO advantage lies between two extremes

INPUT RELATIONSHIPS AND EXCITATION

With nearly independent inputs, SISO contains little shared-input mixing, so the additional MISO coefficient may provide limited benefit. With moderate shared variation and remaining independent information, MISO may separate the pathways more accurately. With extremely high input coherence, the input spectral system can become poorly conditioned and the separated solution can become unstable.

This is a hypothesis about an operating region, not a conclusion. The known-truth simulation is needed because a model that looks plausible or has higher in-sample multiple coherence is not necessarily closer to the true pathways.

SECTION 7

Simulation With a Known Generating Model

The generator creates CBFV using pathway filters that are stored exactly. The estimator receives only finite observed signals and must recover those pathways.

The simulation combines established methods with declared choices

EVIDENCE FOUNDATION

Established methodological core

- spectral simulation of correlated stationary Gaussian processes;
- multiple-input cerebral-hemodynamic models with MAP and respiratory-gas inputs; and
- Welch spectra, joint identification, and known-system validation.

Prespecified study choices

- Gaussian PSDs, flat coherence, amplitudes, and the two pathway forms;
- duration, noise, timing, window, overlap, and ridge conditions; and
- stored-truth SISO–MISO recovery comparisons.

The exact spectra, pathways, and parameter ranges are transparent assumptions to stress-test, not claims that every human record follows this generator.

Foundations: Shinozuka and Deodatis (1991, 1996); Panerai et al. (2000); Peng et al. (2008); Welch (1967); CARNet (2016, 2023, 2024).

Prior work supports the two-input simulation logic

EVIDENCE FOUNDATION

Study	Contribution carried into this design
Panerai et al., 2000	Modeled CBFV jointly from arterial pressure and end-tidal CO_2; used Monte Carlo inputs with matched spectra and examined multivariable identifiability.
Mitsis et al., 2004	Treated pressure and CO_2 as simultaneous dynamic influences and showed why joint pathway structure matters.
Peng et al., 2008	Used multivariate system identification with ABP and respiratory gases, supported by a physiological simulation.
Henley et al., 2015	Connected data-based multivariable identification with a compartmental cerebral-hemodynamic model.
CARNet, 2023–2024	Reinforced current spectral-analysis practice, CO_2 monitoring, validation, and conditioning awareness.

Our contribution is the controlled SISO-versus-MISO recovery map across input coherence, excitation, pathway strength, recording quality, and estimator settings.

A family is a fixed simulated system, not only one signal recording

SIMULATION DESIGN

Each family begins with a fixed relationship between the two inputs and a fixed pair of pathway impulse responses, h_M and h_C. Random MAP and $P_{\mathrm{ET}}CO_2$ signals are generated under those conditions and passed through the pathways to create clean CBFV.

$$V_{\mathrm{clean}}(t) = \big(h_M * M\big)(t) + \big(h_C * C\big)(t).$$

The family therefore represents a generating system. Its long clean signals are random realizations of that system, while the stored impulse responses define the pathway truth that the later estimates are evaluated against.

The test signals are built in nine explicit steps

SIMULATION DESIGN

1. Assign one family's input relationship, pathway strengths, delay, and time constants.
2. Define the target MAP and $P_{\mathrm{ET}}CO_2$ power-spectral shapes.
3. Mix shared and independent random sources to impose target input-input coherence.
4. Transform to time signals, then standardize and scale their fluctuation amplitudes.
5. Construct the true MAP and $P_{\mathrm{ET}}CO_2$ pathway filters.
6. Filter both inputs and sum their contributions to create clean CBFV.
7. Create finite observed records by changing duration, noise, or timing alignment.
8. Estimate SISO and MISO transfer functions from only the observed signals.
9. Compare each estimate with stored truth and test prediction on an independent realization.

The next slides unpack every step. "Known truth" refers to the stored pathway filters, not to a transfer function fitted from the same noisy record.

Step 1: assign the family before drawing any random signal

SIMULATION DESIGN

Each family receives five controlled values: target MAP-$P_{\mathrm{ET}}CO_2$ coherence, spectral similarity, the $P_{\mathrm{ET}}CO_2$/MAP fluctuation-SD ratio, the $P_{\mathrm{ET}}CO_2$/MAP pathway band-gain ratio, and the true $P_{\mathrm{ET}}CO_2$ delay.

Two additional pathway time constants are sampled continuously: 1–4 s for MAP and 2–8 s for $P_{\mathrm{ET}}CO_2$. Together these seven quantities define the generating system. They are fixed before the random MAP and $P_{\mathrm{ET}}CO_2$ sequences are drawn.

The random seed is also stored. Repeating the same family therefore reproduces the same parameter assignment, while changing the training/validation seed creates new fluctuations under the same truth.

Step 2: define where each input has spectral power

SIMULATION DESIGN

The generator first defines smooth broadband power-spectral shapes. MAP uses a Gaussian-shaped spectrum centered at 0.06 Hz with a 0.045-Hz bandwidth. The $P_{\mathrm{ET}}CO_2$ center moves toward or away from the MAP center according to the spectral-similarity control; both spectra also retain a small power floor.

$$P_M(f) = G(f; 0.06, 0.045) + 0.01, \qquad P_C(f) = G(f; f_C, 0.045) + 0.01.$$

This step controls *where power is available*. It does not by itself make the two input waveforms coherent. PSD-shape overlap and input coherence are separate properties.

Spectral-representation basis: Shinozuka and Deodatis (1991), doi:10.1115/1.3119501; Shinozuka and Deodatis (1996), doi:10.1115/1.3101883. The exact Gaussian spectra are study-specific.

Step 3: mix shared and independent sources to set input coherence

SIMULATION DESIGN

Generate two independent white-noise sequences: a shared source Z_s and an independent $P_{\mathrm{ET}}CO_2$ source Z_i. Shape them in the frequency domain:

$$M(f) = \sqrt{P_M(f)}\, Z_s(f),$$
$$C(f) = \sqrt{P_C(f)} \left[\sqrt{\gamma^2_{MC}}\, Z_s(f) + \sqrt{1 - \gamma^2_{MC}}\, Z_i(f) \right].$$

The shared term creates a reproducible MAP-$P_{\mathrm{ET}}CO_2$ relationship; the independent term preserves information unique to $P_{\mathrm{ET}}CO_2$. Because γ^2_{MC} is magnitude-squared coherence, the square-root weights produce the requested coherence in expectation before finite-record estimation error.

Two-channel spectral factorization; see Shinozuka and Deodatis (1996) and Wilson (1972), doi:10.1137/0123044.

Step 4: transform, standardize, and set input amplitudes

SIMULATION DESIGN

Inverse Fourier transforms convert $M(f)$ and $C(f)$ into real time signals. Each generated signal is centered and divided by its own standard deviation. MAP is then assigned an SD of 1 simulation unit, while $P_{\mathrm{ET}}CO_2$ is multiplied by the family's configured $P_{\mathrm{ET}}CO_2$/MAP fluctuation-SD ratio.

$$M(t) \leftarrow \frac{M(t) - \overline{M}}{\mathrm{SD}(M)}, \qquad C(t) \leftarrow r_{\mathrm{SD}} \frac{C(t) - \overline{C}}{\mathrm{SD}(C)}.$$

This makes input excitation explicit: r_{SD} changes how strongly $P_{\mathrm{ET}}CO_2$ fluctuates relative to MAP without changing the separately assigned pathway gain ratio.

Step 5: create and store the two true pathway filters

SIMULATION DESIGN

The MAP pathway is a high-pass-like autoregulatory response: slow MAP changes retain only 25% of the high-frequency gain, while faster changes approach a gain of 1.5. Its shape is governed by the family's MAP time constant.

The $P_{\mathrm{ET}}CO_2$ pathway is a delayed low-pass exponential response governed by its own time constant and the assigned physiological delay. After construction, the complete $P_{\mathrm{ET}}CO_2$ impulse response is scaled so that its RMS gain over 0.005–0.35 Hz has the requested ratio to the MAP pathway's RMS band gain.

Both impulse responses are retained exactly. Their Fourier transforms are the true gain and phase curves used later to score SISO and MISO recovery.

Pathway rationale: Zhang et al. (1998) and CARNet (2016, 2023) for high-pass-like pressure–flow dynamics; Poulin et al. (1996, 1998), Panerai et al. (2000), and Simpson et al. (2000) for delayed CO_2 responses. Exact kernels are study-specific.

Step 6: filter the inputs and add their CBFV contributions

SIMULATION DESIGN

Each input is passed through its own stored pathway filter. The two resulting output components are then added sample by sample:

$$V_M(t) = (h_M * M)(t), \qquad V_C(t) = (h_C * C)(t), \qquad V_{\text{clean}}(t) = V_M(t) + V_C(t).$$

The proof-of-concept uses no nonlinear interaction term. A 120-s burn-in is generated and discarded so initialization transients from the causal filters do not become part of the retained test record. Panel E of the family figure shows $V_M(t)$ and $V_C(t)$; their sum is the clean CBFV shown in panel B.

Step 7: create observed records without changing family truth

SIMULATION DESIGN

The long clean family is converted into multiple finite observations. For each planned condition, the program performs the following operations in order:

1. truncate MAP, $P_{\mathrm{ET}}CO_2$, and CBFV to the requested duration;
2. add MAP and $P_{\mathrm{ET}}CO_2$ measurement noise at their requested input SNRs;
3. add CBFV output noise at the requested output SNR; and
4. apply the requested $P_{\mathrm{ET}}CO_2$ timing misalignment while retaining only the samples shared by all three signals.

The noise sequences are standardized before SNR scaling. Duration, measurement noise, and alignment therefore change what the estimator observes, but the stored h_M and h_C remain the same.

Steps 8–9: estimate from observations, then score against truth

SIMULATION DESIGN

Step 8. Welch's method estimates the auto- and cross-spectra of the observed MAP, $P_{\mathrm{ET}}CO_2$, and CBFV signals. Separate SISO models and the joint two-input MISO model are solved frequency by frequency. Neither estimator is given h_M or h_C.

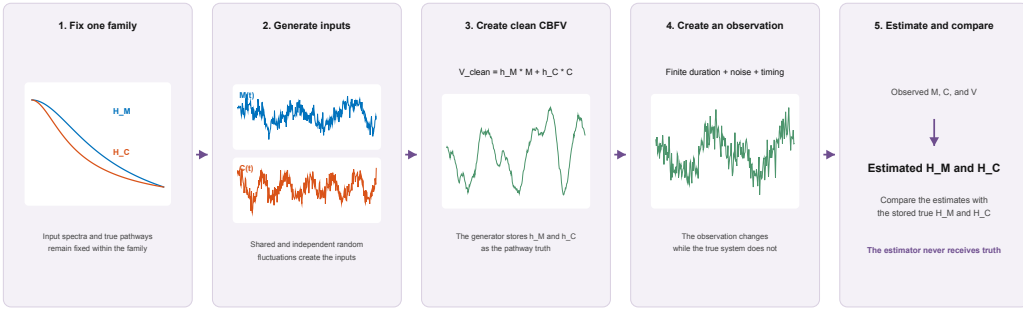
Step 9. The estimated complex transfer functions are compared with the stored true frequency responses. Gain error, wrapped-phase error, and normalized complex error are retained. A second independently generated realization with the same family truth provides an out-of-sample prediction check.

The family remains the independent statistical unit: Welch windows, frequency bins, observations, and estimator settings are repeated measurements of that same generating system.

Estimation follows Welch (1967) and CARNet spectral recommendations; independent-realization validation is consistent with system-identification validation principles emphasized by Kostoglou et al. (2024).

A simulated family moves through five separate stages

SIMULATION DESIGN



One family fixes the generating system; observation and estimator choices test how accurately it can be recovered.

Generated in Python; every displayed trace is a sampled signal realization.

The estimator is not given the true pathways

SIMULATION DESIGN

The generator uses h_M and h_C to create CBFV, but the estimator receives only the observed $M(t)$, $C(t)$, and $V(t)$. From their Welch spectra it produces $\widehat{H}_M(f)$ and $\widehat{H}_C(f)$ as though the signals were experimental recordings.

With infinitely long, noiseless, correctly aligned data and a distinguishable input pair, the correctly specified MISO estimator should approach the generating pathways. The test is meaningful because finite duration, spectral leakage, noise, timing error, and input coherence make the estimated spectra imperfect.

SISO has an additional limitation: it intentionally omits one generating input. When the inputs are related, its pairwise transfer function can differ from the individual pathway even under otherwise favorable conditions.

Training and validation share truth but not random fluctuations

SIMULATION DESIGN

For every family, the program generates two independent baseline realizations using the same family parameters and the same true h_M and h_C. Their random MAP and $P_{\mathrm{ET}}CO_2$ fluctuations differ.

$$\text{fixed family truth} \longrightarrow \begin{cases} \text{training realization: fit } \widehat{H}_M, \widehat{H}_C, \\ \text{validation realization: test prediction on new signals.} \end{cases}$$

Pathway-accuracy outcomes compare the training estimates directly with the stored true frequency responses. The separate validation realization supports an additional out-of-sample prediction check without creating a second family.

A family, observation, and estimator setting are different units

SIMULATION DESIGN

Unit	Meaning in the simulation
Family	One fixed input structure and pair of true MAP and $P_{\mathrm{ET}}CO_2$ pathways, with independent training and validation realizations. The family is the statistical unit.
Observation	A different way of recording the same family: duration, output noise, input measurement noise, or timing alignment may change while truth remains fixed.
Estimator setting	A different analysis of the same observation, such as Welch window length, overlap, or ridge value.

Welch segments, frequency bins, and repeated observations are not counted as independent families.

Read the family figure as three pairs of questions

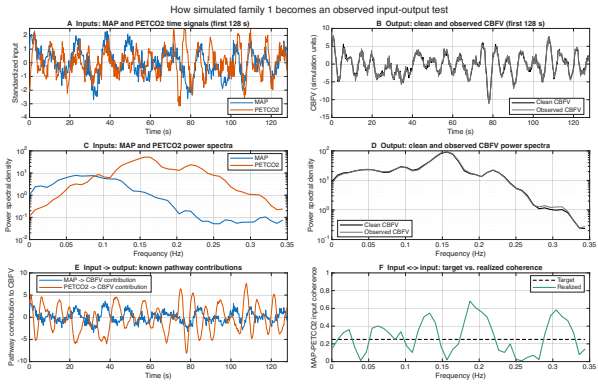
SIMULATION DESIGN

Panels	Relationship	Question answered
A–B	Inputs and output in time	What MAP and $P_{\mathrm{ET}}CO_2$ signals enter the system, and what clean/noisy CBFV signal leaves it?
C–D	Inputs and output in frequency	Where is spectral power available in MAP and $P_{\mathrm{ET}}CO_2$, and where does the resulting CBFV power appear?
E–F	Generating truth and input diagnostic	What did each input pathway contribute to CBFV, and how coherent were the two inputs with each other?

Panels A, C, and F describe the *inputs*; panels B and D describe the *output*; panel E connects each input to its known output contribution. Transfer-function gain and phase are intentionally shown in the next figure, not hidden inside this signal-anatomy figure.

One simulated family shows inputs, output, truth, and input coherence

SIMULATION DESIGN



Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Panels E and F validate different parts of the generator

SIMULATION DESIGN

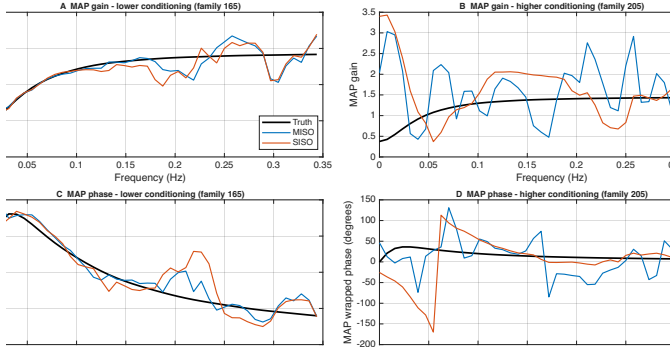
Panel E is input $\rightarrow$ output. It shows the known filtered MAP and $P_{\mathrm{ET}}CO_2$ contributions, $V_M(t)$ and $V_C(t)$. These are not raw inputs and are not estimated transfer functions. Their sample-by-sample sum is clean CBFV:

$$V_{\mathrm{clean}}(t) = V_M(t) + V_C(t).$$

Panel F is input $\leftrightarrow$ input. It compares the configured MAP-$P_{\mathrm{ET}}CO_2$ coherence target with the coherence realized from one finite record. It contains no CBFV and therefore does not measure how well either input explains the output. Variation around the dashed target is expected because the solid curve is a finite-record Welch estimate.

MAP → CBFV gain and phase are evaluated against truth

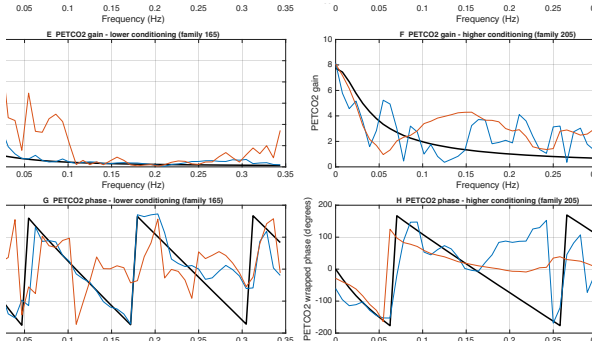
SIMULATION DESIGN



Top row: MAP → CBFV gain. Bottom row: MAP → CBFV phase. Black is truth; blue is MISO; orange is SISO. Columns compare lower and higher conditioning.

$P_{\mathrm{ET}}CO_2 \rightarrow$ CBFV gain and phase are evaluated against truth

SIMULATION DESIGN



Top row: $P_{\mathrm{ET}}CO_2 \rightarrow$ CBFV gain. Bottom row: $P_{\mathrm{ET}}CO_2 \rightarrow$ CBFV phase. Black is truth; blue is MISO; orange is SISO. Columns compare lower and higher conditioning.

Five intrinsic controls define the proof-of-concept families

SIMULATION DESIGN

Assigned family factor	Five configured values
Target MAP-$P_{ET}CO_2$ coherence	0.05, 0.25, 0.50, 0.75, 0.95
Spectral-similarity control	0, 0.25, 0.50, 0.75, 1
$P_{ET}CO_2$/MAP fluctuation SD ratio	0.05, 0.25, 0.50, 1, 2
$P_{ET}CO_2$/MAP pathway band-gain ratio	0.10, 0.50, 1, 2, 4
True $P_{ET}CO_2$ pathway delay	0, 2.5, 5, 7.5, 10 s

MAP and $P_{ET}CO_2$ pathway time constants remain continuously sampled nuisance properties over 1–4 s and 2–8 s. They are not forced into categories only to make the figures look uniform.

Pairwise balance replaces an unnecessary full factorial

SIMULATION DESIGN

A complete five-factor design at five levels would require

$$5^5 = 3{,}125 \text{ families.}$$

The proof design instead uses a 25-family base block in which each factor level appears five times and every pair of factors contains all 5×5 level combinations once. Ten independently generated blocks give

$$25 \text{ families per block} \times 10 \text{ blocks} = 250 \text{ families.}$$

Therefore each assigned level has 50 families and each assigned pairwise cell has 10. The design protects planned one-factor and two-factor summaries without claiming to cover all 3,125 five-factor combinations.

Assigned factors and realized characteristics must remain distinct

SIMULATION DESIGN

Assigned factors are selected before signal generation and are displayed at their exact configured values. Realized characteristics are measured from the longest clean generated record. They can differ slightly from their targets and are continuous rather than five-valued by definition.

For presentation operating maps, realized input coherence, PSD-shape overlap, and $P_{\mathrm{ET}}CO_2$ contribution-power share use five fixed intervals:

$$[0, 0.2),\ [0.2, 0.4),\ [0.4, 0.6),\ [0.6, 0.8),\ [0.8, 1].$$

These boundaries were specified before the 250-family results were inspected and were not moved afterward merely to fill heatmap cells.

Every family receives the same paired observation plan

SIMULATION DESIGN

Experiment	Levels applied to every family
Duration by CBFV SNR	Seven durations (256–896 s) crossed with seven output-SNR levels (∞ to 0 dB): 49 observations
MAP measurement noise	Seven input-SNR levels at the reference observation
$P_{\mathrm{ET}}CO_2$ measurement noise	Seven input-SNR levels at the reference observation
$P_{\mathrm{ET}}CO_2$ timing misalignment	$-6, -3, 0, 3, 6$ s at the reference observation

The reference observation is 300 s with 15-dB CBFV output SNR, no added input measurement noise, and no timing misalignment. The plan contains 68 primary observations per family, producing $250 \times 68 = 17{,}000$ primary analyses.

Estimator sensitivity is repeated within each family

SIMULATION DESIGN

The primary estimator uses a 128-s Hann window, 50% overlap, and the common three-point smoothing kernel [0.25, 0.50, 0.25]. At least three complete Welch windows are required.

Sensitivity analyses evaluate ridge values 0.001, 0.01, and 0.1; Welch windows of 64, 75, 100, 128, and 150 s; overlaps of 25%, 50%, and 75%; and the full duration-by-window grid. This adds 750 ridge analyses and 10,750 estimator-setting analyses to the 250-family run.

Because these are repeated analyses of the same underlying families, they increase the number of computations without increasing the independent sample size beyond 250.

The simulation code mirrors the generating equations

CODE IMPLEMENTATION

```
for family = 1:numberOfFamilies
    [hM,hC] = createTruePathways(family);
    [M,C] = generateRelatedInputs(family);

    Vclean = filter(hM,M) + filter(hC,C);
    [Mobs,Cobs,Vobs] = applyObservationPlan( ...
        M,C,Vclean,duration,noise,timing);

    spectra = welchSpectra(Mobs,Cobs,Vobs);
    Hhat = solveMISO(spectra);
    error = compareWithTruth(Hhat,hM,hC);
end
```

Conceptual implementation of the simulation workflow. The production project separates these operations into tested generator, observation, spectral-estimation, solver, and metric functions.

SECTION 8

How the Estimates Are Compared With Truth

The primary outcome compares the complete complex transfer function, while gain and phase errors explain which part of the estimate changed.

Complex error combines magnitude and phase disagreement

KNOWN-TRUTH OUTCOMES

For an estimated pathway $\widehat{H}(f)$ and its known truth $H(f)$, the code averages squared complex distance over the analyzed frequency range and normalizes by the mean energy of the true transfer function:

$$E_{\text{complex}} = \frac{\text{mean}_f \left| \widehat{H}(f) - H(f) \right|^2}{\text{mean}_f \, |H(f)|^2}.$$

This retains disagreement in both gain and phase. Separate absolute gain and wrapped-phase errors are also calculated to make the result easier to interpret.

Phase error is evaluated only where the true pathway is defined

KNOWN-TRUTH OUTCOMES

Phase becomes unstable or physically uninformative where true pathway gain is nearly zero. The implementation therefore evaluates wrapped phase error only at frequencies where true gain exceeds 5% of its root-mean-square gain.

$$e_{\phi}(f) = \angle\left[\exp\left(i\{\angle\widehat{H}(f) - \angle H(f)\}\right)\right].$$

This returns the principal angular difference rather than treating a 2π representation change as a large physiological error.

Model advantage expresses the relative error on a log scale

KNOWN-TRUTH OUTCOMES

For the same family and observation, SISO and MISO error are converted into a model-advantage value:

$$A_{\mathrm{model}} = \log_{10}\left(\frac{E_{\mathrm{SISO}} + \epsilon}{E_{\mathrm{MISO}} + \epsilon}\right), \qquad \epsilon = 10^{-12}.$$

$A_{\mathrm{model}} < 0$: **SISO has lower error** $A_{\mathrm{model}} = 0$: equal error $A_{\mathrm{model}} > 0$: **MISO has lower error**

$A_{\mathrm{model}} = 0.30$ means the SISO error is about twice the MISO error; $A_{\mathrm{model}} = 1$ means it is ten times larger. Always show absolute SISO and MISO errors too: a favorable ratio can coexist with poor performance by both models.

Aggregation preserves the family as the independent unit

KNOWN-TRUTH OUTCOMES

Each displayed group or heatmap cell is summarized across families, not across Welch segments or frequency bins treated as independent repetitions. Exported source data retain the mean, standard deviation, valid family count, total group count, standard error, confidence interval, and probability values when inference is performed.

The presentation display threshold is five valid families. Cells below this threshold remain gray and receive no mean color. Five families permit a descriptive calculation; they do not by themselves provide strong publication evidence. Frequency-resolved plots keep the native Welch frequency bins rather than rebinned frequencies.

SECTION 9

Audited 250-Family Proof-of-Concept Results

The run verifies the design and demonstrates the expected MISO accuracy–stability tradeoff. These results guide the final paper design rather than replace it.

The proof-of-concept run completed as designed

SIMULATION AUDIT

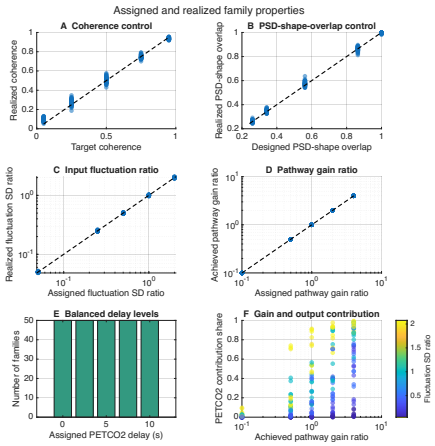
Completed quantity	Count
Independent simulated families	250
Complete 25-family balanced blocks	10
Primary observation analyses	17,000
Ridge analyses	750
Estimator-setting analyses	10,750
Audited metric-condition support checks	748
Numerical runtime	21.3 min

All 14 core execution checks passed. All 25 marginal level checks and all 250 assigned pairwise-cell checks passed. No audited whole-range metric value was silently discarded as nonfinite.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Assigned controls were reproduced by the generator

GENERATOR VALIDATION



Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Balanced checkpoints support using 250 families for this deck

SIMULATION AUDIT

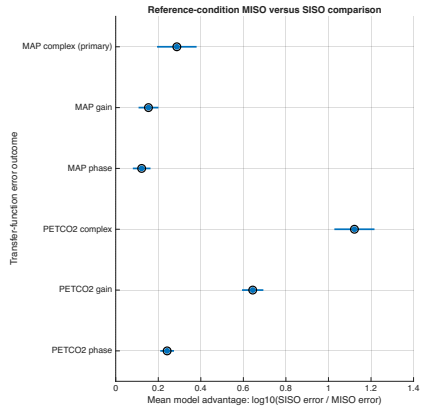
The design contains natural complete-block checkpoints at 125, 200, and 250 families. Across the six reference-condition complex, gain, and phase model-advantage outcomes, the largest absolute difference from the final 250-family mean was 0.0339 at 125 families and 0.0169 at 200 families.

These are descriptive stability checks rather than additional hypothesis tests. They show that the main proof-of-concept averages were not being driven by the last few families, while still leaving the larger 1,000-family profile necessary for stronger cell support and final publication-level modeling.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

At the reference condition, MISO had lower mean error

REFERENCE-CONDITION RESULT



Means and 95% confidence intervals across 250 families. Black rings mark the reported comparison after the stated adjustment family. Model advantage is $\log_{10}(E_{\mathrm{SISO}} / E_{\mathrm{MISO}})$.

The reference result was larger for the $P_{\mathrm{ET}}CO_2$ pathway

REFERENCE-CONDITION RESULT

At 300 s, 15-dB CBFV output SNR, and no input noise or timing shift, mean MAP complex model advantage was 0.287 (95% CI 0.194 to 0.380). This corresponds to an approximately 1.94-fold geometric SISO-to-MISO error ratio.

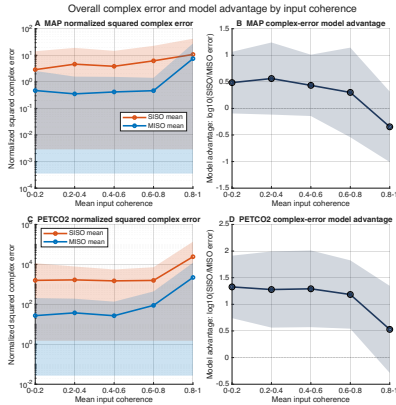
Mean $P_{\mathrm{ET}}CO_2$ complex model advantage was 1.122 (95% CI 1.028 to 1.216), corresponding to an approximately 13.25-fold geometric error ratio. Gain and phase comparisons were also positive on average for both pathways.

This establishes a proof-of-concept average difference, not a claim that MISO is better in every family. The next analyses examine where that average changes.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Very high input coherence changes the MAP tradeoff

INPUT IDENTIFIABILITY RESULT



Families are grouped by prespecified intervals of realized mean input coherence. Curves show across-family means; shaded regions show SD.

The highest-coherence group illustrates the stability limit

INPUT IDENTIFIABILITY RESULT

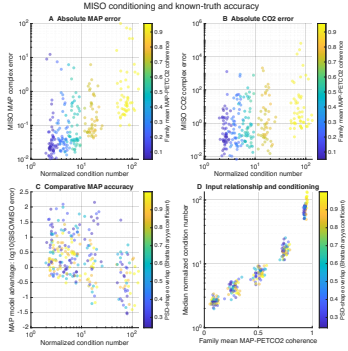
MAP model advantage remained positive through the 0.6–0.8 realized-coherence group, but the 0.8–1.0 group had a mean advantage of −0.346 (95% CI −0.534 to −0.157). Its geometric SISO-to-MISO error ratio was 0.45, indicating lower MAP SISO error on average in that group.

The $P_{\mathrm{ET}}CO_2$ advantage also decreased at the highest coherence but remained positive on average: 0.529 (95% CI 0.297 to 0.762). This difference between pathways is important. Poor identifiability does not have to affect both true pathways in the same way because their excitation and pathway gains also differ.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Condition number connects coherence with known-truth error

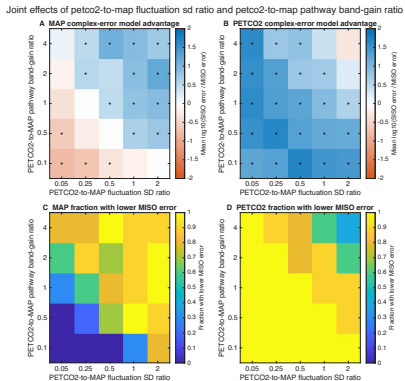
INPUT IDENTIFIABILITY RESULT



Each point represents one family at the reference observation. Conditioning is calculated after scaling the inputs to unit spectral power.

Excitation and pathway gain must be interpreted together

PATHWAY EXCITATION RESULT



Axes use exact assigned values; each 5×5 pairwise cell contains 10 families. Black dots mark adjusted $P < 0.05$.

Weak $P_{\mathrm{ET}}CO_2$ influence can limit the MAP benefit

PATHWAY EXCITATION RESULT

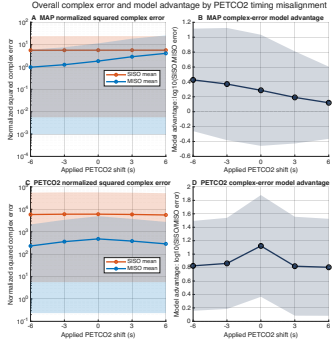
The joint assigned-factor map separates the strength of $P_{\mathrm{ET}}CO_2$ fluctuations from the strength of its pathway to CBFV. MAP MISO advantage is weakest, and can be negative, when both the $P_{\mathrm{ET}}CO_2$ input and its pathway are very weak. In that region there is little second-pathway influence for MISO to separate, so an additional coefficient can add finite-record variability without removing much SISO mixing.

As either the $P_{\mathrm{ET}}CO_2$ fluctuations or pathway gain becomes stronger, MAP MISO advantage generally increases. The $P_{\mathrm{ET}}CO_2$ pathway shows a broader positive MISO advantage, but its exact magnitude also changes across the joint design. This supports interpreting excitation and pathway strength as separate controls rather than one combined label.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Timing alignment changes the two pathways differently

OBSERVATION-QUALITY RESULT



Each timing shift is applied to the same 250 families at the reference duration and output SNR. Curves show across-family means; shaded regions show SD.

The timing result is directional rather than a single penalty

OBSERVATION-QUALITY RESULT

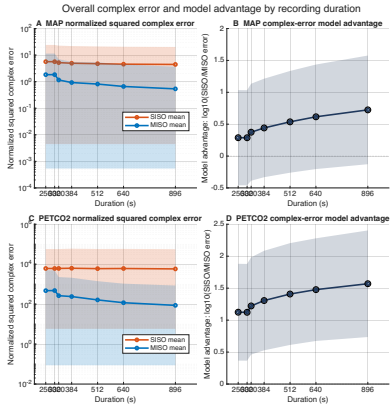
The $P_{\mathrm{ET}}CO_2$ complex model advantage was largest with no applied timing shift (1.122) and decreased to approximately 0.80–0.86 after shifts of three or six seconds in either direction. MAP advantage changed asymmetrically: it was 0.287 at zero shift, 0.120 at $+6$ s, and 0.427 at -6 s.

This asymmetry is consistent with timing error interacting with the families' true $P_{\mathrm{ET}}CO_2$ delays rather than acting as one uniform amount of noise. A recording's alignment choice can therefore change which complex pathway is easier to recover and should be treated as part of the model specification.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

Longer records generally strengthen the advantage

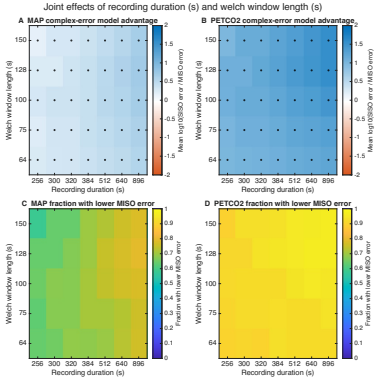
FINITE-RECORD RESULT



Durations use exact configured values. All curves use 15-dB CBFV output SNR and the primary 128-s, 50%-overlap Welch estimator.

Duration and Welch window length form one estimator tradeoff

FINITE-RECORD RESULT



Axes use exact configured values. Each cell summarizes the same 250 families; black dots mark adjusted $P < 0.05$.

The interaction map supports an intuitive interpretation

FINITE-RECORD RESULT

Longer recordings generally increase the amount of information available to both models and increase the fraction of families in which MISO has lower complex error. Window length modifies this result because it controls the balance between frequency resolution and the number of segment-level estimates that can be averaged.

The $P_{\mathrm{ET}}CO_2$ pathway shows a broad MISO advantage across the tested duration-window grid. The MAP pathway shows more gradual improvement with duration and greater dependence on window choice. This difference reinforces that estimator settings should not be selected only from one pathway's behavior.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

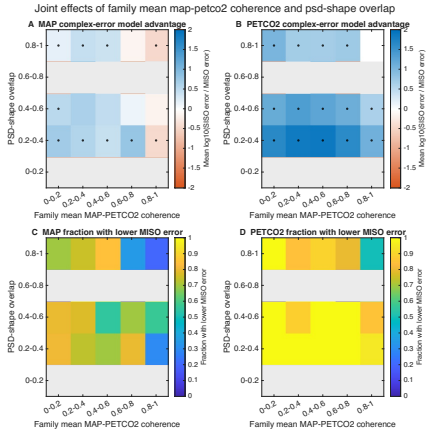
SECTION 10

What the Proof of Concept Establishes–and What It Does Not

The run supports the workflow and the main tradeoff, while also identifying design decisions that must be made before the final paper simulation.

Realized bins reveal where the current design is sparse

PROOF-OF-CONCEPT LIMITATIONS



Gray cells contain fewer than five valid families. The fixed realized-value boundaries were not moved after seeing occupancy.

The gray cells are information about support, not missing results

PROOF-OF-CONCEPT LIMITATIONS

Realized coherence occupied the five bins almost evenly: 51, 49, 50, 50, and 50 families. Realized PSD-shape overlap was uneven: 0, 100, 45, 5, and 100. Across the three audited two-dimensional realized-characteristic grids, 43 of 75 cells met the five-family display threshold and 13 were empty.

The unsupported cells remain gray because the proof design did not generate enough families there. Filling them by changing bin boundaries after seeing the data would make the figure look more complete without adding information. These maps are therefore descriptive and should not be treated as precise boundaries of an operating region.

Evidence status: `proof250`, 250-family presentation proof of concept; not final-paper evidence.

The 1,000-family run is a planned extension, not a simple rerun

TRANSITION TO THE FINAL PAPER

Repeating the 25-family balanced block 40 times would provide 1,000 families, 40 families per assigned pairwise cell, and greater support for continuous models or finer descriptive grouping. Before that run, the following choices should be specified prospectively:

- final continuous-bin boundaries and minimum cell size;
- which interactions are primary, secondary, or supplemental;
- whether spectral-similarity coverage should be expanded;
- the primary endpoint and multiplicity families; and
- which family-level continuous regression models will support formal claims.

The larger sample should increase evidence, not replace the design decisions revealed by the proof of concept.

The current conclusion is methodological and conditional

PROOF-OF-CONCEPT SYNTHESIS

The 250-family run shows that MISO can recover both known pathways more accurately on average under the reference observation, with an especially large average advantage for the $P_{\mathrm{ET}}CO_2$ pathway. It also shows why that conclusion cannot be reduced to "MISO is better." MAP accuracy can favor SISO when the inputs are extremely coherent, and MISO benefit changes with excitation, pathway gain, timing, record duration, and estimator settings.

The next scientific step is to finalize the paper-scale design using these audit findings, run the 1,000-family profile, and only then place the cognitively normal human recordings within the supported simulated operating space.

SECTION A

Methods and Reference Appendix

The essential equations belong in the paper's main Methods. The expanded derivations, assumptions, and source bank are retained here to stage the supplement and make every design choice traceable.

The paper and supplement should overlap where the math carries meaning

APPENDIX ROAD MAP

Keep visible in the main paper

- the 2×2 spectral model;
- the shared/independent input construction;
- both true pathways and the observation plan; and
- complex error plus signed model advantage.

Expand in the supplement

- SISO contamination and the explicit MISO solution;
- achieved-coherence and conditioning derivations;
- PSD, scaling, burn-in, seed, SNR, and trimming details; and
- phase masks, validation, sensitivity grids, and aggregation rules.

Some equations should appear twice: once because they are necessary to understand the main method, and again as the starting point for reproduction.

The input generator is a two-channel spectral factorization

MATHEMATICAL FOUNDATION

For the study's zero-cross-phase construction, prescribe the target input spectral matrix

$$S_{xx}(f) = \begin{bmatrix} P_M(f) & \sqrt{P_M(f)P_C(f)\gamma^2(f)} \\ \sqrt{P_M(f)P_C(f)\gamma^2(f)} & P_C(f) \end{bmatrix}.$$

This positive-semidefinite matrix can be factored as $S_{xx} = LL^T$. Multiplying independent complex Gaussian Fourier coefficients by $L(f)$ produces correlated coefficients with the requested second-order spectra. Conjugate symmetry across positive and negative frequencies yields real time signals after the inverse FFT.

Spectral-representation foundation: Shinozuka and Deodatis (1991, 1996); matrix factorization: Wilson (1972).

The factor has a direct shared-plus-independent interpretation

MATHEMATICAL FOUNDATION

Let $Z_s(f)$ and $Z_i(f)$ be independent, unit-power complex Gaussian sources. One valid lower-triangular factorization is

$$\begin{bmatrix} M(f) \\ C(f) \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{P_M(f)} & 0 \\ \sqrt{P_C(f)\gamma^2(f)} & \sqrt{P_C(f)[1-\gamma^2(f)]} \end{bmatrix}}_{\mathsf{L}(f)} \begin{bmatrix} Z_s(f) \\ Z_i(f) \end{bmatrix}.$$

The first column is shared by MAP and $P_{\mathrm{ET}}CO_2$; the second contributes only to $P_{\mathrm{ET}}CO_2$. As $\gamma^2 \to 1$, unique $P_{\mathrm{ET}}CO_2$ information disappears. As $\gamma^2 \to 0$, the input fluctuations become independent in expectation.

The construction achieves the requested coherence in expectation

MATHEMATICAL FOUNDATION

Using $E|Z_s|^2 = E|Z_i|^2 = 1$ and $E[Z_s^* Z_i] = 0$ gives

$$S_{MM} = P_M, \qquad S_{CC} = P_C, \qquad S_{MC} = \sqrt{P_M P_C \gamma^2}.$$

Therefore

$$\frac{|S_{MC}(f)|^2}{S_{MM}(f) S_{CC}(f)} = \frac{P_M(f) P_C(f) \gamma^2(f)}{P_M(f) P_C(f)} = \gamma^2(f).$$

The configured $\gamma^2(f)$ is flat and has zero cross-spectral phase. A finite Welch estimate fluctuates around that target because only a limited number of segments are averaged. This difference between configured and realized coherence is expected, not generator failure.

PSD overlap and coherence remain mathematically distinct

MATHEMATICAL FOUNDATION

PSD-shape overlap depends only on the two marginal spectral shapes. One possible normalized overlap summary is

$$O_{MC} = \frac{\int P_M(f) P_C(f)\, df}{\sqrt{\int P_M^2(f)\, df \int P_C^2(f)\, df}}.$$

Input coherence instead depends on their cross-spectrum:

$$\gamma^2_{MC}(f) = \frac{|S_{MC}(f)|^2}{S_{MM}(f) S_{CC}(f)}.$$

Two signals may have similar PSD shapes but independent fluctuations (O_{MC} high, γ^2_{MC} low), or may be coherent only within a narrow frequency range. Excitation adds a third question: whether either input has enough power in the analyzed band to identify its pathway.

The stored MAP truth is high-pass-like by construction

TRUE PATHWAY CONSTRUCTION

Let $L_M(f)$ be the unit-gain exponential low-pass response, G_M the target high-frequency gain, and r_M the retained low-frequency fraction. The code constructs

$$H_M(f) = G_M\left[r_M + (1 - r_M)\{1 - L_M(f)\}\right] = G_M\left[1 - (1 - r_M)L_M(f)\right].$$

Because $L_M(0) \approx 1$, slow MAP changes have gain $G_M r_M$. As frequency increases and $L_M(f) \to 0$, the gain approaches G_M. In the proof design, $r_M = 0.25$ and $G_M = 1.5$ simulation units; the family-specific time constant controls the transition. This is a transparent test kernel, not a fitted human autoregulation model.

Physiological shape rationale: Zhang et al. (1998); Claassen et al. (2016); Panerai et al. (2023).

The stored $P_{\mathrm{ET}}CO_2$ truth is delayed and low-pass

TRUE PATHWAY CONSTRUCTION

Before finite impulse-response truncation, the conceptual pathway is

$$h_C(t) = G_C \frac{1}{\tau_C} e^{-(t-d_C)/\tau_C} u(t-d_C), \qquad H_C(f) = G_C \frac{e^{-i2\pi f d_C}}{1 + i2\pi f \tau_C}.$$

The implementation samples and normalizes the causal exponential, prepends an integer-sample delay, and truncates after a specified number of time constants. It then rescales the complete kernel so its RMS gain over 0.005–0.35 Hz has the assigned ratio to the MAP pathway's RMS band gain. Thus delay, time constant, and pathway strength are controlled separately.

Delayed CO_2 response rationale: Poulin et al. (1996, 1998); Panerai et al. (2000); Simpson et al. (2000).

The observation model changes the data, not the stored truth

OBSERVATION MODEL

The generating system first creates

$$V_0(t) = (h_M * M)(t) + (h_C * C)(t).$$

An observation then adds independent measurement sequences scaled to the requested signal-to-noise ratios:

$$M_o = M + n_M, \qquad C_o = C + n_C, \qquad V_o = V_0 + n_V, \qquad P_n = P_s 10^{-\mathrm{SNR_{dB}}/10}.$$

A nonzero alignment condition shifts the observed $P_{\mathrm{ET}}CO_2$ index relative to MAP and CBFV, then trims all signals to their common support. In frequency-domain terms, an ideal time shift multiplies the $P_{\mathrm{ET}}CO_2$ coefficient by $e^{\pm i2\pi f\Delta}$, directly changing cross-spectral phase. None of these operations alters stored h_M or h_C.

The SISO mixing term follows from one omitted input

ESTIMATOR DERIVATION

Start from $V = H_M M + H_C C + E$, with E uncorrelated with both inputs. Multiplying by M^* and taking expectation gives

$$S_{MV} = E[M^* V] = H_M S_{MM} + H_C S_{MC}.$$

Dividing by S_{MM} gives

$$H_M^{\mathrm{SISO}} = \frac{S_{MV}}{S_{MM}} = H_M + H_C \frac{S_{MC}}{S_{MM}}.$$

The additional term vanishes when $S_{MC} = 0$, but otherwise inherits the frequency-dependent magnitude and phase of both the $P_{\mathrm{ET}}CO_2$ pathway and the input relationship. This is omitted-input mixing, not an assertion that the MAP pairwise transfer function was computed incorrectly.

The 2×2 solve separates both coefficients simultaneously

ESTIMATOR DERIVATION

$$\begin{bmatrix} S_{MM} & S_{MC} \\ S_{CM} & S_{CC} \end{bmatrix} \begin{bmatrix} H_M \\ H_C \end{bmatrix} = \begin{bmatrix} S_{MV} \\ S_{CV} \end{bmatrix}, \qquad D = S_{MM} S_{CC} - |S_{MC}|^2.$$

For $D \neq 0$,

$$H_M = \frac{S_{CC} S_{MV} - S_{MC} S_{CV}}{D}, \qquad H_C = \frac{S_{MM} S_{CV} - S_{CM} S_{MV}}{D}.$$

The main paper should retain the matrix equation because it is the shortest clear definition of the estimator. The explicit formulas belong in the supplement because they expose the common denominator and the source of finite-record instability.

Coherence gives a closed-form conditioning warning

ESTIMATOR DERIVATION

After scaling both inputs to unit spectral power, the input matrix is

$$\widetilde{S}_{xx} = \begin{bmatrix} 1 & \rho \\ \rho^* & 1 \end{bmatrix}, \qquad |\rho|^2 = \gamma_{MC}^2.$$

Its eigenvalues are $\lambda_{\max} = 1 + |\rho|$ and $\lambda_{\min} = 1 - |\rho|$, so the two-norm condition number is

$$\kappa_2(\widetilde{S}_{xx}) = \frac{1 + |\rho|}{1 - |\rho|} = \frac{1 + \sqrt{\gamma_{MC}^2}}{1 - \sqrt{\gamma_{MC}^2}}.$$

As coherence approaches one, the normalized determinant approaches zero and the condition number diverges. This formalizes the tradeoff: shared variation creates SISO mixing, while too little independent variation destabilizes MISO.

Truth scoring keeps absolute error and relative advantage separate

ESTIMATOR DERIVATION

For either pathway,

$$E_{\text{complex}} = \frac{\text{mean}_f\,|\widehat{H}(f) - H(f)|^2}{\text{mean}_f\,|H(f)|^2}, \qquad A_{\text{model}} = \log_{10}\frac{E_{\text{SISO}} + \epsilon}{E_{\text{MISO}} + \epsilon}.$$

$A_{\text{model}} < 0$: **SISO is better** $A_{\text{model}} = 0$: equal $A_{\text{model}} > 0$: **MISO is better**

Because $10^{A_{\text{model}}} = (E_{\text{SISO}} + \epsilon)/(E_{\text{MISO}} + \epsilon)$, 0.301 corresponds to approximately twofold lower MISO error and -0.301 to approximately twofold lower SISO error. Absolute errors must accompany the ratio so that a relative advantage cannot hide poor recovery by both models.

The proof-of-concept generator makes five simplifying assumptions

SCOPE OF THE MATHEMATICAL MODEL

- Input realizations are stationary and Gaussian.
- MAP and $P_{\mathrm{ET}}CO_2$ act through additive, linear, time-invariant pathways.
- Target input coherence is flat and has zero cross-spectral phase.
- PSD shapes, causal pathway kernels, and parameter ranges are specified by the study.
- Pathway truth is stored exactly while duration, noise, and timing alter only the observation.

These restrictions are useful because they isolate estimator behavior. They must remain explicit wherever simulation validity is discussed.

Those assumptions limit the physiological claims we can make

SCOPE OF THE MATHEMATICAL MODEL

- The proof does not show that resting human inputs are exactly Gaussian or stationary.
- It does not rule out nonlinearity, feedback, unmeasured inputs, or time variation.
- It does not imply that physiological input coherence is flat or phase-free.
- The selected spectra, pathways, and parameter ranges are not population distributions.
- Human pathway truth remains unobservable and cannot be validated as it is here.

External validity therefore depends on sensitivity analyses and on locating human recordings within the supported simulated operating space.

Foundational spectral-analysis and TFA sources

REFERENCE BANK

- Welch PD. The use of fast Fourier transform for the estimation of power spectra. *IEEE Trans Audio Electroacoust*. 1967;15:70–73. doi:10.1109/TAU.1967.1161901.
- Zhang R, Zuckerman JH, Giller CA, Levine BD. Transfer function analysis of dynamic cerebral autoregulation in humans. *Am J Physiol*. 1998;274:H233–H241. doi:10.1152/ajpheart.1998.274.1.H233.
- Claassen JAHR et al. Transfer function analysis of dynamic cerebral autoregulation: a white paper from the International Cerebral Autoregulation Research Network. *J Cereb Blood Flow Metab*. 2016;36:665–680. doi:10.1177/0271678X15626425.

These sources define the core frequency-domain estimator and the cerebral-TFA interpretation used in the main Methods.

Current CARNet recommendations retained for reporting

REFERENCE BANK

- Panerai RB et al. Transfer function analysis of dynamic cerebral autoregulation: a CARNet white paper 2022 update. *J Cereb Blood Flow Metab*. 2023;43:3–25. doi:10.1177/0271678X221119760.
- Kostoglou K et al. Time-domain methods for quantifying dynamic cerebral blood flow autoregulation: review and recommendations. *J Cereb Blood Flow Metab*. 2024;44:1480–1514. doi:10.1177/0271678X241249276.

The 2023 paper is the direct current TFA standard. The 2024 paper is retained for its broader treatment of multiple inputs, validation, correlation, and conditioning; it is not presented as direct endorsement of this frequency-domain MISO implementation.

Direct multivariable cerebral-hemodynamic sources

REFERENCE BANK

- Panerai RB, Simpson DM, Deverson ST, Mahony P, Hayes P, Evans DH. Multivariate dynamic analysis of cerebral blood flow regulation in humans. *IEEE Trans Biomed Eng*. 2000;47:419–423. doi:10.1109/10.827312.
- Peng T, Rowley AB, Ainslie PN, Poulin MJ, Payne SJ. Multivariate system identification for cerebral autoregulation. *Ann Biomed Eng*. 2008;36:308–320. doi:10.1007/s10439-007-9412-9.
- Mitsis GD, Poulin MJ, Robbins PA, Marmarelis VZ. Nonlinear modeling of the dynamic effects of arterial pressure and CO_2 variations on cerebral blood flow in healthy humans. *IEEE Trans Biomed Eng*. 2004;51:1932–1943. doi:10.1109/TBME.2004.834272.
- Henley BC et al. Compartmental and data-based modeling of cerebral hemodynamics: linear analysis. *IEEE Access*. 2015;3:2317–2332. doi:10.1109/ACCESS.2015.2492945; PMCID: PMC4756910.
- Marmarelis VZ et al. Multiple-input nonlinear modelling of cerebral haemodynamics using spontaneous arterial blood pressure, end-tidal CO_2 and heart rate measurements. *Phil Trans R Soc A*. 2016;374:20150180. doi:10.1098/rsta.2015.0180.
- Panerai RB, Eames PJ, Potter JF. Multiple coherence of cerebral blood flow velocity in humans. *Am J Physiol Heart Circ Physiol*. 2006;291:H251–H259. doi:10.1152/ajpheart.01348.2005.

Pathway-physiology sources retained for manuscript development

REFERENCE BANK

- Panerai RB, Deverson ST, Mahony P, Hayes P, Evans DH. Effects of CO_2 on dynamic cerebral autoregulation measurement. *Physiol Meas*. 1999;20:265–275. doi:10.1088/0967-3334/20/3/304.
- Poulin MJ, Liang PJ, Robbins PA. Dynamics of the cerebral blood flow response to step changes in end-tidal PCO_2 and PO_2 in humans. *J Appl Physiol*. 1996;81:1084–1095. doi:10.1152/jappl.1996.81.3.1084.
- Poulin MJ et al. Fast and slow components of cerebral blood flow response to step decreases in end-tidal PCO_2 in humans. *J Appl Physiol*. 1998;85:388–397. doi:10.1152/jappl.1998.85.2.388.
- Simpson DM, Panerai RB, Evans DH, Garnham J, Naylor AR, Bell PR. Estimating normal and pathological dynamic responses in cerebral blood flow velocity to step changes in end-tidal pCO_2. *Med Biol Eng Comput*. 2000;38:535–539. doi:10.1007/BF02345749.
- Panerai RB, Dawson SL, Potter JF. Linear and nonlinear analysis of human dynamic cerebral autoregulation. *Am J Physiol Heart Circ Physiol*. 1999;277:H1089–H1099. doi:10.1152/ajpheart.1999.277.3.H1089.

These papers motivate pathway form and sensitivity analyses; they do not make the exact proof-of-concept kernels uniquely physiological.

Stochastic-simulation and spectral-factorization sources

REFERENCE BANK

- Shinozuka M, Deodatis G. Simulation of stochastic processes by spectral representation. *Appl Mech Rev.* 1991;44:191–204. doi:10.1115/1.3119501.
- Shinozuka M, Deodatis G. Simulation of multi-dimensional Gaussian stochastic fields by spectral representation. *Appl Mech Rev.* 1996;49:29–53. doi:10.1115/1.3101883.
- Wilson GT. The factorization of matricial spectral densities. *SIAM J Appl Math.* 1972;23:420–426. doi:10.1137/0123044.

Together these sources support the general method of specifying marginal and cross-spectra, factorizing the spectral matrix, generating independent random Fourier coefficients, and transforming them into correlated stationary Gaussian signals. Our two-source formula is the closed-form factorization for the particular 2×2, zero-cross-phase matrix used here.

The accompanying source ledger records each citation's claim, intended deck and manuscript use, evidence status, and limitation so that references considered during development are not lost even when they are not cited in the main text.

REVIEW STOPPING POINT

The narrative, mathematical foundation, and source bank are now connected.

The next review should decide which appendix equations move verbatim into the manuscript, which implementation details remain supplemental, and whether any literature-backed sensitivity conditions should be added before the final 1,000-family simulation specification.